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Narrow Brown Leaf Spot of Rice

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Narrow brown leaf spot (NBLS), also referred to as *Cercospora* leaf spot (CLS), *Cercospora* blotch (CNB) or *Cercospora* panicle blight (CPB), is a widespread foliar disease of rice caused by *Cercospora janseana* (syn. *Cercospora oryzae*, *Sphaerulina oryzina*). It occurs throughout tropical and subtropical rice-growing regions and has emerged as a production constraint in the southern United States, especially in ratoon rice systems. NBLS can reduce yield of up to 40% and impair grain quality, and contribute to plant lodging.

Symptoms and Signs





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early senescence.

- **Sheaths:** Net-like patterns of alternating brown and yellow areas, termed Cercospora net blotch.
- **Panicles and Glumes:** Lesions may extend to panicle branches and glumes in severe cases, causing grain discoloration and reduced milling quality.
- **Other Signs:** Under humid conditions, sporulation of the pathogen may be observed as fine, dark fungal growth on lesion surfaces.

Pathogen Biology

Cercospora janseana is a fungal pathogen with a genome size of 39,075,509 base pairs, distributed across 15 chromosomes. The pathogen is seedborne, enabling long-distance spread, and it also survives in infected crop residue and ratoon stubble. It produces multiseptated, geniculate conidiophores ($80\text{--}140 \times 4\text{--}9 \mu\text{m}$), which emerge from stomata. Conidia are cylindrical to club-shaped ($15\text{--}60 \times 3\text{--}5 \mu\text{m}$), hyaline to light olive, with 3 to 10 septa, and are borne in whorls. The pathogen disperses via wind, rain splash, or mechanical activity. It is polycyclic and thrives at 25 to 28°C with a pH range of 5.7 to 7.1. Multiple races of the pathogen are commonly present and can overcome resistance in cultivars over time.

Disease Cycle

The pathogen survives in seed, infected crop debris, and ratoon stubble. Under favorable environmental conditions, conidia (asexual spores) are produced and dispersed via wind, rain splash, or mechanical means. Primary infections arise when conidia infect lower leaves, entering through stomata and growing intercellularly. The disease subsequently progresses upward in the canopy, with conidia produced on infected tissue promoting secondary infection cycles. *Sphaerulina oryzina* is the teleomorph of *C. janseana*; however, its role in the disease cycle remains uncertain, as it is rarely observed in the field. Recent population genetic studies suggest a potential contribution of sexual reproduction to pathogen diversity. Ratoon (second crops) are particularly vulnerable due to regrowth from infected stubble and inoculum carryover from the main crop.

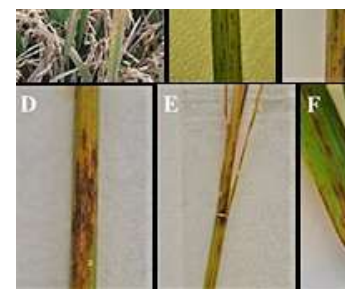


Figure 1: Symptoms and signs of Narrow Brown Leaf Spot (NBLS) caused by *Cercospora janseana*.

- A. Severe lesions causing leaf death.
- B. Narrow, linear brown lesions (0.5–1 mm) on rice leaves aligned with the veins.
- C. Linear brown lesions on the sheath.
- D. Net blotch appearance on the upper sheath, a diagnostic feature of NBLS.
- E. Panicle branches with brown lesions indicative of *Cercospora* panicle blight.
- F. Discoloration of glumes and compromised grain quality in severely infected samples.

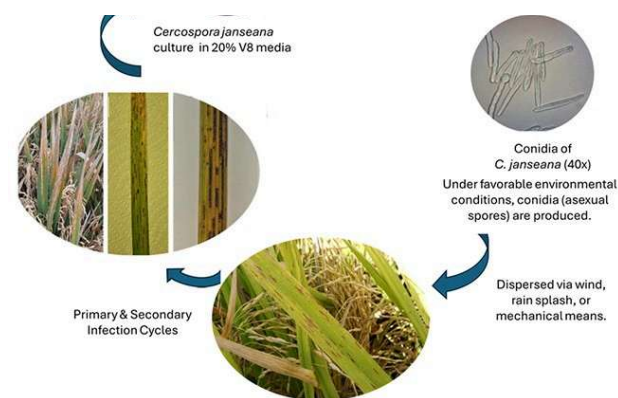

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Figure 2: Disease cycle of Narrow Brown Leaf Spot (NBLS) of rice caused by *Cercospora janseana*.

infection, though they may take up to 30 days to develop depending on weather conditions, pathogen race, rice cultivar susceptibility, and the plant's growth stage.

- **Susceptibility:** All growth stages are susceptible, peak vulnerability from heading to maturity.
- **Environment:** High humidity, warm temperature to 28 °C), frequent rainfall, and potassium deficiency increase disease severity.
- **Cropping System:** Continuous rice cultivation and ratoon (second) cropping tend to exhibit higher disease incidence due to residual inoculum buildup

the field.

- **Cultural Factors:** Late planting, dense planting, low levels of nitrogen, and poor drainage exacerbate NBLS.

Management

Host Resistance

- Plant resistant or moderately resistant cultivars based on local recommendations. Hybrid cultivars generally exhibit greater resistance than inbred cultivars. Cultivars with high levels of resistance are available, and a major resistance gene, CRSP-2.1, has been identified contributing to this trait.
- Resistance ratings for specific regions are available in the rice disease management guide of states, including Arkansas, Louisiana and Texas.

Fungicide Application

- Apply fungicides from the boot to heading stages for the main crop; apply at the boot stage for the ratoon crop. Two fungicide applications are recommended for susceptible cultivars under highly conducive conditions.
- Effective fungicides include:
 - **Propiconazole-based:** Tilt[®] [propiconazole (FRAC Code 3)], PropiMax[®] (propiconazole), Stratego[®] [trifloxystrobin (FRAC Code 11) + propiconazole]
 - **Combination products:** Quilt[®] (azoxystrobin + propiconazole), QuiltXcel[®] [azoxystrobin (FRAC Code 11) + propiconazole], Amistar Top[®] [azoxystrobin +

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- In multi-year field trials, **propiconazole, azoxystrobin, and pyraclostrobin** consistently reduced NBLS severity by up to 85% and improved grain quality.

Cultural Practices

- Rotate with non-host crops to reduce fungal inoculum buildup.
- Remove infected residue after harvest.
- Manage grassy weeds such as *Panicum repens* L. (alternate host) and volunteer rice.
- Ensure balanced nutrition with sufficient nitrogen and potassium fertilization.
- Ensure proper irrigation management to reduce leaf wetness, limiting fungal spread.
- Optimize planting density to reduce humidity in the canopy.

Summary

NBLS has become a recurring challenge in U.S. rice production, especially in ratoon cropping systems where inoculum carryover is common. Field research indicates that well-timed fungicide applications — especially at the heading stage for the main crop and the boot stage for the ratoon — can significantly reduce disease severity. An integrated disease management approach is essential for maintaining rice yield, grain quality, and the long-term sustainable production.

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